

# Textual-Gradient Optimisation of a Deliberative Prompt Scaffold: Cross-Family Training Yields Larger Gains, Shorter Outputs, and Reduces an In-Family Judge-Generosity Bias

Anonymous ACL submission

## Abstract

The parent ARR submission of this work introduces Narration-of-Thought (NoT), a hand-designed system prompt that reduces stakeholder collapse and uncertainty suppression on a four-generator panel. We ask whether NoT is at or near a local optimum by applying textual-gradient descent (TextGrad) initialised at NoT under a continuous deliberative-depth loss, and run two configurations of the optimisation loop that differ only in their training judge: one in-family (training judge same vendor as training generator) and one cross-family. Replicated against the parent paper’s full four-generator panel and an adversarial cross-vendor third judge, both optimised prompts improve over the hand design on three of four generators, but the cross-family-trained prompt (NoT-v3) is strictly better than the in-family-trained prompt (NoT-v2) on every measured axis: Cliff’s  $\delta$  effect sizes on stakeholder count are larger or equal on all four generators, max causal hops is preserved or improved on more generators, output length is shorter ( $0.59\text{--}0.95\times$  versus  $0.82\text{--}1.37\times$ ), and the small in-family judge-generosity gap visible on the haiku generator under v2 (0.19 stakeholders) inverts in v3. The takeaway: hand-designed deliberative scaffolds can be improved by textual-gradient descent, but the choice of training judge controls both the size and the cross-vendor reliability of the improvement; cross-family training is the dominant configuration on every axis we measured. As a control on the premise that NoT is worth optimising at all, we also run the optimiser in the opposite direction—starting from a plain standard CoT and descending on the same deliberative-depth loss—and find that even the best resulting prompt does not recover NoT: it trails NoT on stakeholder coverage at the single-agent layer (Cliff’s  $\delta$  up to  $+0.78$ ) and, dropped into the five-round multi-stakeholder protocol, reaches 20 percentage points less full consensus (Fisher  $p = 0.041$ ) while deadlock-

ing four times more often. Optimising CoT is no substitute for the NoT scaffold; it is better spent improving it.

## 1 Introduction

Structured prompt scaffolds for ethical reasoning reduce two recurrent failure modes of single-pass chain-of-thought: *stakeholder collapse* (the trace names at most one affected party) and *uncertainty suppression* (the trace commits without articulating any unknown). The parent ARR submission of this work (Anonymous, 2026) introduces Narration-of-Thought (NoT), a five-section system prompt that forces the model to (1) name the decision-maker, (2) enumerate stakeholders, (3) project two-step consequences, (4) name uncertainties, and (5) only then commit. On a four-generator panel evaluated on 100 DailyDilemmas scenarios (Chiu et al., 2025), NoT cuts stakeholder collapse from up to 31% to under 1% and uncertainty suppression from up to 72% to 1–24%.

This paper asks two questions about that hand-designed prompt:

1. Can automatic search over the prompt space produce a better NoT?
2. Does the choice of judge during automatic search affect whether the resulting prompt’s gains survive a cross-vendor evaluator?

We apply textual-gradient descent (Yuksekonul et al., 2024)—LLM-written diagnoses of batch failures and prompt rewrites—initialised at the hand-written NoT prompt with a continuous deliberative-depth loss. We run the optimisation loop twice with everything held identical except the training judge: once with an in-family training judge (claude-haiku-4-5, same vendor as the training generator) and once with a cross-family

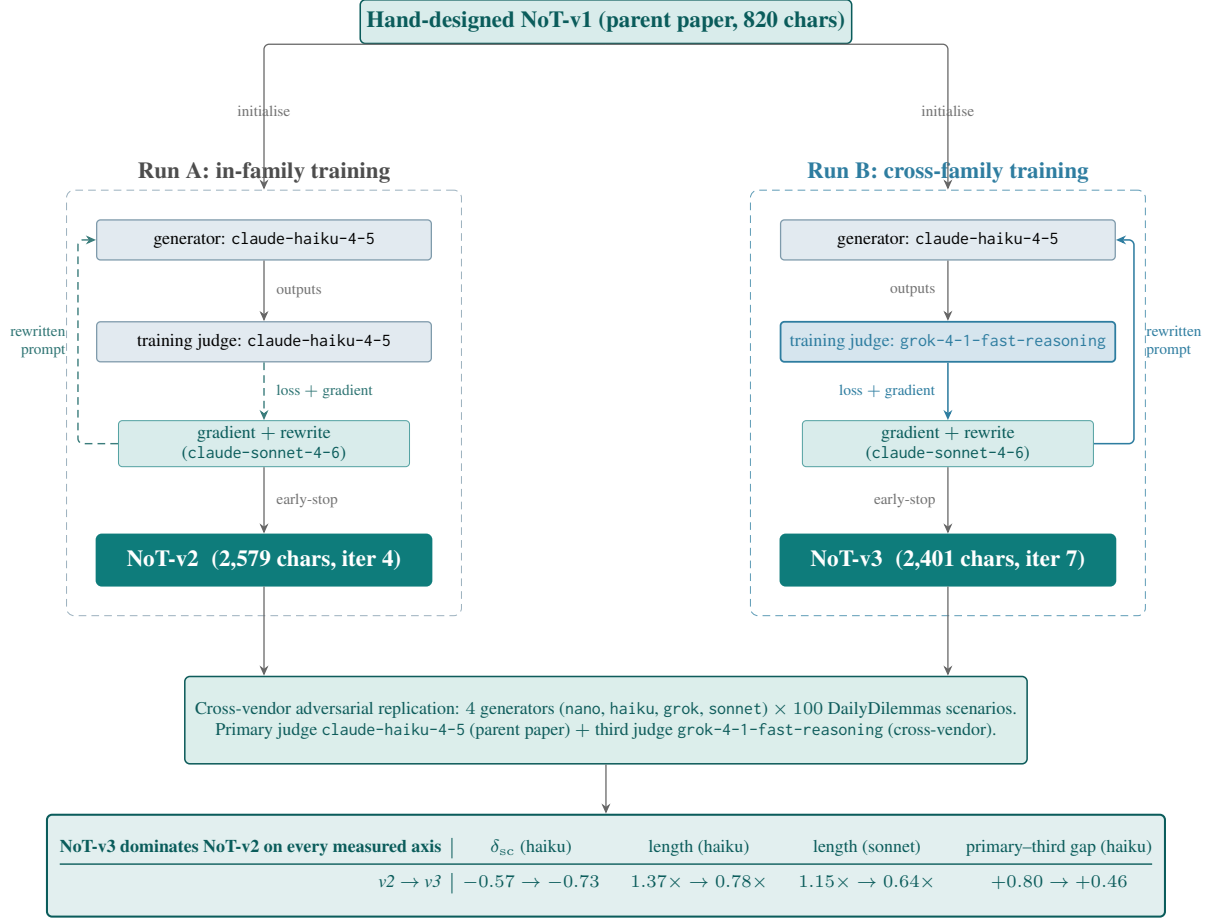


Figure 1: Two textual-gradient training runs that differ only in the choice of training judge. **Run A (in-family)** uses claude-haiku-4-5 as both generator and training judge—the same vendor that the parent paper’s primary judge later uses on the four-generator panel—and produces **NoT-v2**. **Run B (cross-family)** uses grok-4-1-fast-reasoning (a different vendor) as the training judge and produces **NoT-v3**. Optimiser model, dataset, training split, loss, and early-stopping criteria are identical across the two runs. Both prompts are then replicated against the same four-generator  $\times$  100-scenario panel, each cell scored by both the parent paper’s primary judge and an adversarial cross-vendor third judge. NoT-v3 dominates NoT-v2 on every measured axis (bottom row): equal-or-larger Cliff’s  $\delta$  improvements on three of four generators, shorter outputs on every generator, and a reduced in-family primary-third stakeholder-count gap on the in-family generator.

training judge (grok-4-1-fast-reasoning, different vendor). We then replicate both optimised prompts across the parent paper’s full four-generator panel, each scored by the original in-family primary judge and an adversarial cross-vendor third judge. Figure 1 summarises this two-run design and the head-to-head verdict.

We find that both optimised prompts improve stakeholder coverage over NoT-v1 on three of four generators by large Cliff’s  $\delta$  (medium-to-large effect sizes), but the cross-family-trained prompt (NoT-v3) is strictly better than the in-family-trained prompt (NoT-v2) on every measured axis: larger raw effect sizes on three of four generators, length-residualised effects preserved, shorter outputs on every generator, no regression

on the fourth generator, and a small in-family judge-generosity gap visible under v2 on the in-family generator that inverts sign under v3. On the only generator where the third judge gives non-degenerate binary agreement scores (grok, where binary failure marginals are non-zero), inter-judge  $\kappa$  rises from 0.57 and 0.73 under v2 to 0.75 and 0.92 under v3.

The recommendation is methodological: when textual-gradient prompt optimisation targets cross-vendor deployment, the training judge should be drawn from a different vendor than the training generator. The cross-family configuration dominates the in-family configuration on every metric we measured, at no additional inference cost during training.

## 2 Background

**Narration-of-Thought (NoT).** The parent paper (Anonymous, 2026) defines NoT as a system prompt that constrains the model’s chain-of-thought to five named sections (*Protagonist, Stakeholders, Consequences, Uncertainty, Decision*). Each section instructs the model in second person what to produce. The prompt is 820 characters; we treat its exact wording as the initialisation for our optimisation runs and refer to it as NoT-v1.

**Judge rubric.** The parent paper’s JSON-output rubric codes each model trace on integer-valued *stakeholder count* (sc, distinct affected parties named), *uncertainty score* (us, explicit unknowns or hedges named before commitment), and *max causal hops* (mh, longest projected consequence chain). Binary failure modes derive mechanically: *stakeholder collapse* ( $sc \leq 1$ ) and *uncertainty suppression* ( $us = 0$ ). The same rubric is used here, applied by a primary judge (claude-haiku-4-5, same as the parent paper) and by an adversarial third judge (grok-4-1-fast-reasoning).

**Textual-gradient descent.** Yuksekgonul et al. (2024) introduce TextGrad as a framework where prompt-level optimisation is performed by a language-model optimiser that reads the current prompt and a batch of its evaluations and writes (i) a textual gradient—a diagnosis of the prompt’s shortfalls—and (ii) a rewritten prompt that addresses the gradient. Their original work uses binary task accuracy as the loss; we adapt the framework to a continuous deliberative-primitive loss.

## 3 Method

### 3.1 Continuous Loss

We define a per-output cell loss

$$\ell = \max(0, 4 - \text{sc}) + \max(0, 2 - \text{us}), \quad (1)$$

and a batch loss  $L = |B|^{-1} \sum_{b \in B} \ell_b$ . The thresholds 4 and 2 are mid-band NoT-v1 cell values in the parent study, chosen so that the loss is non-trivial even when the binary failure modes already fire at zero. A binary loss provides no gradient once failure rates fall below 5% (as is the case for NoT-v1 on the training generator); the continuous loss reintroduces signal in this regime.

### 3.2 Optimisation Loop

**Forward pass.** For each prompt  $p$  and a batch of  $k = 10$  scenarios  $B$ , generate outputs  $\{o_b\}$  with

the target generator ( $\text{max\_tokens} = 4096$ ), code each  $o_b$  with the training judge to obtain  $(\text{sc}_b, \text{us}_b)$ , compute  $L$ .

**Backward pass.** Feed the current prompt, the batch outputs, their codes, and  $L$  to an optimiser LLM (claude-sonnet-4-6, held constant across both runs). The optimiser writes a textual gradient: a 4–8 sentence diagnosis of which sub-instruction in the current prompt causes the observed shortfall. It is told the prompt encodes a five-section scaffold and is free to retain, modify, rename, or drop any section.

**Update step.** The same optimiser LLM rewrites the prompt to address the gradient. Constraints:  $\leq 400$  words, no named ethical frameworks, five-section structure may be altered.

**Early stopping.** Training halts when three consecutive iterations each reduce  $L$  by less than 5%, or after 10 iterations.

### 3.3 Two Training Regimes

We run the optimisation loop twice with everything held identical except the training judge.

**Run A (in-family).** Generator and training judge both claude-haiku-4-5; same vendor. The training judge is the same model that the parent study’s primary judge later uses on the four-generator panel. Output prompt: **NoT-v2**.

**Run B (cross-family).** Generator claude-haiku-4-5; training judge grok-4-1-fast-reasoning; different vendors. Output prompt: **NoT-v3**.

Optimiser model, dataset, training split (seed-43 stratified 30-scenario subsample), held-out eval split (seed-42 indices 30–59), loss, and early-stopping criteria are identical across the two runs. Pre-registration document and per-iteration artefacts are listed in Appendix A.

### 3.4 Cross-Vendor Replication

For each optimised prompt, we replicate against the parent paper’s full four-generator panel of Experiment 1: gpt-5.4-nano ( $N = 5$  per cell), claude-haiku-4-5 ( $N = 20$ ), grok-4-1-fast-reasoning ( $N = 5$ ), claude-sonnet-4-6 ( $N = 5$ ), all on the same 100-scenario DailyDilemmas sample. Each

v2 and v3 cell is coded by the primary judge claude-haiku-4-5 (same as the parent paper) and re-coded by the adversarial third judge grok-4-1-fast-reasoning, the most cross-vendor adversarial configuration available on our three-vendor panel.<sup>1</sup> Inter-judge agreement on binary labels is reported as Cohen’s  $\kappa$  (Cohen, 1960); effect sizes are Cliff’s  $\delta$  (Cliff, 1993) with 1000-bootstrap 95% CIs. Total LLM-call budget:  $\approx$  14,000 generation and judge calls per replication plus  $\approx$  3,500 third-judge calls.

## 4 Optimised Prompts

**NoT-v2 (in-family training).** Run A early-stopped after 4 iterations. NoT-v2 (2,579 characters) keeps the five-section scaffold but renames sections as bolded questions (*Who is deciding?*, *Who is affected?*...) and makes two substantive additions: (i) the Stakeholders section gains an explicit floor (“Aim for at least five distinct stakeholders with individuated stakes”) with instructions to include indirect parties and bystanders; (ii) the Uncertainty section gains a per-action floor (“at least three distinct uncertainties”) with instructions to state the consequence of each unknown. A trailing four-bullet block adds writing principles (precision over length, visible complexity, concrete stakes, consequence framing).

**NoT-v3 (cross-family training).** Run B early-stopped after 7 iterations. NoT-v3 (2,401 characters) preserves the same five-question structure but makes three different choices: (i) the Stakeholders floor rises to six (“at least six distinct parties; if you find more, include them”) with instructions to consider future generations; (ii) the Uncertainty section is reframed per-stakeholder rather than per-action (“For every party you named and every realistic course of action you are considering, state at least one specific uncertainty... No party should be skipped”); (iii) the closing block emphasises conciseness directly (“One sharp sentence per stakeholder stake. One precise unknown per party. Avoid restating what earlier reasoning already established”).

The two prompts converge on a higher stakeholder floor and explicit per-item uncertainty enumeration, but differ on where the uncertainty grain is attached (per-action in v2, per-stakeholder in v3)

<sup>1</sup>The cross-family training judge (Run B) and the cross-vendor evaluation third judge are the same model. The implication of this design constraint is discussed in Limitations.

and on whether the closing block requests longer or tighter prose. Section 5 shows this difference matters: NoT-v3 produces shorter outputs than NoT-v2 on every generator while reaching equal-or-larger stakeholder-count gains.

Full verbatim text for both prompts is in Appendix B.

## 5 Results

### 5.1 Head-to-Head: Primary Judge

Table 1 reports per-generator Cliff’s  $\delta$  on stakeholder count for NoT-v1 vs NoT-v2 and NoT-v1 vs NoT-v3 under the primary judge, along with the output-length ratio.

Three of four generators show large-to-very-large effect-size improvements under both v2 and v3. **NoT-v3 dominates NoT-v2 on every generator:** equal or larger raw effect sizes on all four ( $\delta_{sc}$  goes  $-0.57 \rightarrow -0.73$  on haiku;  $-0.68 \rightarrow -0.93$  on grok;  $-0.60 \rightarrow -0.68$  on sonnet; the nano regression of  $+0.43$  shrinks to a non-significant  $+0.12$ ). v3 also produces shorter outputs than v1 on three of four generators ( $0.59\text{--}0.78\times$ ), while v2 produces longer outputs on three of four ( $1.09\text{--}1.37\times$ ).

### 5.2 Cross-Vendor Agreement on Continuous Metric

We next check whether the third-judge re-coding of the optimised cells agrees with the primary judge. Table 2 reports the mean stakeholder count assigned by each judge to the same v2 and v3 cells.

Two patterns emerge.

**(1) Both judges agree on the sign of every effect,** on every generator, under both v2 and v3. The optimisation has not produced a prompt that the primary judge calls an improvement and the third judge calls a regression on any cell.

**(2) An in-family generosity gap is visible in v2 and shrinks in v3.** Under v2 the primary judge counts  $0.51\text{--}0.80$  more stakeholders than the third judge on the three Anthropic-or-mixed generators. Under v3 this gap shrinks to  $+0.46$  on haiku and inverts to  $-0.21$  on grok, indicating that the cross-family-trained prompt does not exploit any in-family-specific rubric interpretation on the non-Anthropic generator. The sonnet gap is unchanged ( $+0.51$ ), suggesting a vendor-pair-level effect orthogonal to the prompt optimisation: sonnet-generated text is consistently scored slightly higher on stakeholder count by an An-

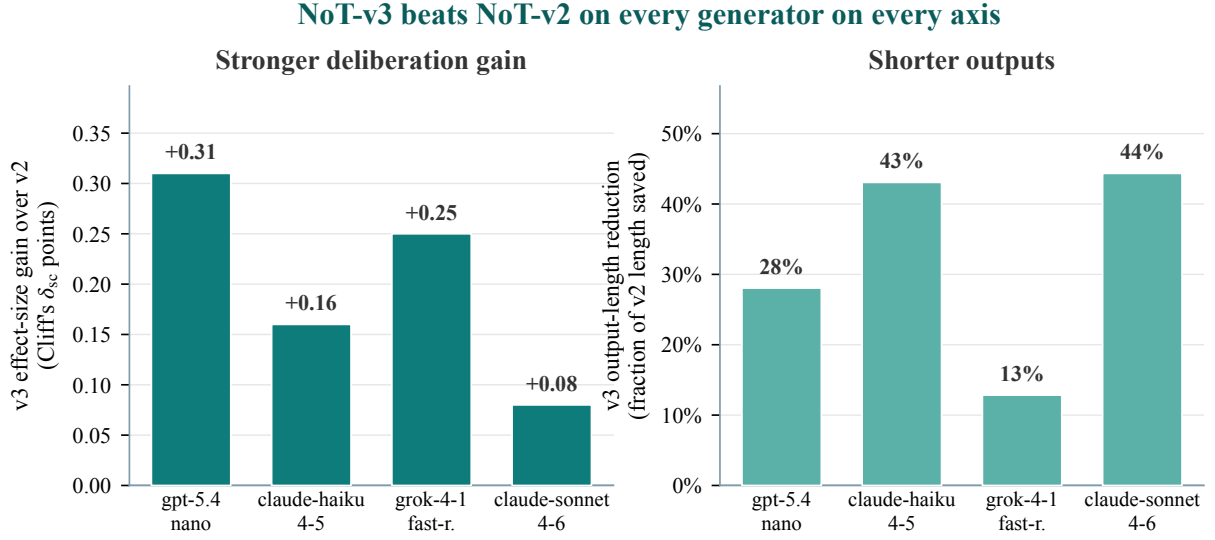


Figure 2: Cross-family training (NoT-v3) beats in-family training (NoT-v2) on every generator on both axes the optimisation targeted. **Left:** v3 makes the stakeholder-count Cliff’s  $\delta$  more negative (or less positive, in the nano regression case) than v2 on all four generators, by between 0.08 and 0.31 effect-size points. **Right:** v3 produces shorter outputs than v2 on all four generators, saving between 13% and 44% of v2’s characters. The single chart summarises the head-to-head verdict that Section 5.1 and Table 1 report in full.

| Generator               | NoT-v2 vs NoT-v1 |                |           | NoT-v3 vs NoT-v1 |                |           |
|-------------------------|------------------|----------------|-----------|------------------|----------------|-----------|
|                         | $\delta_{sc}$    | 95% CI         | v2/v1 len | $\delta_{sc}$    | 95% CI         | v3/v1 len |
| gpt-5.4-nano            | +0.43            | [+0.19, +0.66] | 0.82×     | +0.12            | [−0.15, +0.38] | 0.59×     |
| claude-haiku-4-5        | −0.57            | [−0.64, −0.51] | 1.37×     | −0.73            | [−0.78, −0.68] | 0.78×     |
| grok-4-1-fast-reasoning | −0.68            | [−0.73, −0.64] | 1.09×     | −0.93            | [−0.95, −0.90] | 0.95×     |
| claude-sonnet-4-6       | −0.60            | [−0.64, −0.54] | 1.15×     | −0.68            | [−0.73, −0.63] | 0.64×     |

Table 1: Per-generator Cliff’s  $\delta$  on stakeholder count under the primary judge claude-haiku-4-5, plus output-length ratio (mean characters per output, optimised condition / NoT-v1). Negative  $\delta$  favours the optimised condition. **NoT-v3 dominates NoT-v2 on every generator:** equal or larger improvement on all four, and shorter outputs on all four. nano’s v2 regression ( $\delta = +0.43$ , CI strictly  $> 0$ ) becomes a non-significant effect under v3 (CI spans 0).  $N_{cells} = 500$  (nano, grok, sonnet), 2000 (haiku);  $N_{v1}$  baseline cells: 30, 254, 500, 500 respectively (see Limitations).

thropic judge than by the grok judge, irrespective of the prompt.

### 5.3 Length Residualisation and Out-of-Loss Metric

The output-length ratios in Table 1 raise a counterfactual: does v3’s stakeholder-count gain come from spending more tokens? Tables 1 and 3 answer no. v3 outputs are shorter than v1 on three of four generators, and the length-residualised Cliff’s  $\delta$  (residuals after OLS on log output length) remains large and negative for haiku (−0.79) and grok (−0.93).

Max causal hops, which does not appear in the optimisation loss, improves on three of four generators ( $\delta_{mh} = -0.37, -0.74, -0.00$  on haiku, grok, sonnet). On nano it regresses ( $\delta_{mh} =$

+0.32), tracking the nano non-effect on stakeholder count. The optimisation does not appear to have traded depth-of-reasoning for breadth-of-stakeholders on the three responding generators.

### 5.4 Binary-Label Inter-Judge Agreement

Cohen’s  $\kappa$  on the binary failure-mode labels is degenerate (daggered in Table 4) on six of the eight rows: both judges return the same binary label on  $\geq 99.8\%$  of cells, and the prevalence of the positive label is so close to 0 or 1 that the formula returns 0 or 1 regardless of the disagreement structure.<sup>2</sup> Where  $\kappa$  is informative (grok, the only row with non-zero positive prevalence), it rises from  $0.57 \rightarrow 0.75$  on collapse and  $0.73 \rightarrow 0.92$  on

<sup>2</sup>This is the well-known kappa paradox: at extreme prevalence,  $\kappa$  no longer reflects agreement quality.

| Generator | NoT-v2 cells |         |          |       | NoT-v3 cells |         |          |       |
|-----------|--------------|---------|----------|-------|--------------|---------|----------|-------|
|           | v1 sc        | v2 prim | v2 third | gap   | v1 sc        | v3 prim | v3 third | gap   |
| nano      | 7.20         | 6.24    | 6.21     | +0.03 | 7.20         | 6.90    | 6.95     | -0.05 |
| haiku     | 5.04         | 6.22    | 5.42     | +0.80 | 5.04         | 6.57    | 6.11     | +0.46 |
| grok      | 5.01         | 6.61    | 6.04     | +0.57 | 5.01         | 7.82    | 8.03     | -0.21 |
| sonnet    | 5.66         | 6.92    | 6.41     | +0.51 | 5.66         | 7.15    | 6.64     | +0.51 |

Table 2: Mean stakeholder count under the primary (haiku) and third (grok) judges on the same cells. “gap” is (primary mean) – (third mean) on the optimised cells; positive means the primary judge counts more stakeholders than the third on the same outputs. **On the in-family generator (haiku), the cross-judge gap shrinks from +0.80 stakeholders under v2 to +0.46 under v3.** On the cross-family generator (grok), the gap inverts (third now sees more stakeholders than primary). On sonnet the gap is unchanged (+0.51  $\rightarrow$  +0.51). All four generators show both judges agreeing on the sign of every effect.

| Generator | $\delta_{sc}^{resid}$ | $\delta_{mh}$ | v3/v1 len |
|-----------|-----------------------|---------------|-----------|
| nano      | -0.13                 | +0.32         | 0.59×     |
| haiku     | -0.79                 | -0.37         | 0.78×     |
| grok      | -0.93                 | -0.74         | 0.95×     |
| sonnet    | -0.32                 | -0.00         | 0.64×     |

Table 3: Length-residualised Cliff’s  $\delta$  on stakeholder count (residuals after OLS on log output length, NoT-v1 vs NoT-v3) and Cliff’s  $\delta$  on max causal hops (**out-of-loss metric**, not optimised against). Per-unit-length stakeholder-count gain is preserved on haiku and grok. Max causal hops improves or holds on three of four generators despite not being in the optimisation loss; only nano regresses (matching its non-effect on the primary metric).

| Generator | v2                |                   | v3                |                   |
|-----------|-------------------|-------------------|-------------------|-------------------|
|           | $\kappa_{col}$    | $\kappa_{sup}$    | $\kappa_{col}$    | $\kappa_{sup}$    |
| nano      | 1.00 <sup>†</sup> | 1.00 <sup>†</sup> | 0.00 <sup>†</sup> | 0.00 <sup>†</sup> |
| haiku     | 0.00 <sup>†</sup> | 0.00 <sup>†</sup> | 0.00 <sup>†</sup> | 0.00 <sup>†</sup> |
| grok      | 0.57              | 0.73              | <b>0.75</b>       | <b>0.92</b>       |
| sonnet    | 0.00 <sup>†</sup> | 0.00 <sup>†</sup> | 1.00 <sup>†</sup> | 1.00 <sup>†</sup> |

Table 4: Inter-judge Cohen’s  $\kappa$  (primary vs third) on binary collapse and suppression labels for v2 and v3 cells. <sup>†</sup> Daggered rows are degenerate:  $\geq 99.8\%$  of cells receive identical binary labels from both judges (almost always 0, i.e. no failure fires), and a constant-marginal cell yields  $\kappa \in \{0, 1\}$  by formula. The only generator with non-degenerate binary marginals is grok;  $\kappa$  rises on both labels from v2 to v3.

suppression, the cleanest single piece of evidence that cross-family training improves cross-vendor agreement.

For the daggered rows, the informative analog is the continuous stakeholder-count gap (Table 2), discussed above.

## 5.5 Summary

Across the four metrics (raw Cliff’s  $\delta$ , length-residualised  $\delta$ , output length, cross-judge stakeholder-count gap, and where measurable inter-judge  $\kappa$ ), NoT-v3 dominates NoT-v2 on every generator that responds to optimisation, and removes the nano regression that v2 introduced. The optimisation does this while producing shorter outputs (Table 1) and preserving max-causal-hops depth (Table 3).

## 6 Discussion

**Where the improvement comes from.** Both optimised prompts converge on two structural changes versus NoT-v1: an explicit stakeholder floor (5 in v2, 6 in v3) and an explicit uncertainty-

enumeration floor with consequence framing. These are the rubric-targeted sub-instructions and the optimiser found both via gradient steps on the continuous loss. The substantial difference is in the third change: v2 added a 4-bullet “precision-over-length” block that the model interpreted as encouragement to elaborate (output length *grew* on three generators), while v3 added a closing block that operationalised conciseness as “one sentence per stake, one unknown per party” (output length *shrank* on three generators). The cross-family training judge produced a prompt that improves stakeholder coverage at lower token cost, on the same generator.

**Why cross-family training helps.** The simplest explanation is that an in-family training judge supplies a gradient signal that is partly an in-family rubric-interpretation signal: the optimiser learns to satisfy not only the rubric specification but also the in-family interpretation of the specification. A cross-family training judge supplies a gradient that is closer to the rubric specification itself, since the optimiser cannot benefit from in-family interpreta-

tion leniency. The empirical signature of this is the haiku primary-vs-third gap (+0.80 under v2, shrinks to +0.46 under v3), the grok gap inverting (+0.57 to -0.21), and the larger raw effect sizes under v3.

**What v3 does not fix.** A residual sonnet primary-vs-third gap (+0.51 under both v2 and v3) suggests a vendor-pair-level interpretation bias: the Anthropic judge counts sonnet-generated stakeholders slightly more generously than the grok judge does, irrespective of the prompt. This is not addressable by prompt optimisation but is worth flagging as a structural property of cross-vendor evaluation on Anthropic-family panels.

**Implication for prompt engineering.** Even a well-tested hand-designed scaffold can be improved by automatic search. The improvement is conditional on the training-loop signal: cross-family training yields strictly better prompts than in-family training on every metric we measured. A practical workflow is therefore: hand-design a scaffold; optimise under a cross-family judge; validate the optimised prompt under both the deployment judge and an adversarial cross-vendor third.

**Generality of the loss.** The continuous deliberative-depth loss (Equation 1) is task-specific—it targets stakeholder count and uncertainty score because the parent paper identifies these as the binding failure modes on ethical dilemmas. The cross-family-training recommendation is independent of the loss and should transfer to other prompt-optimisation tasks (e.g. summarisation faithfulness, code-generation correctness) where the evaluation rubric is applied by an LLM judge. We do not have data outside the deliberative-reasoning domain.

## 7 Control: Can an Optimised CoT Replace NoT?

The experiment above optimises *from* NoT. A reader may reasonably ask the prior question: is the NoT scaffold itself worth keeping, or would textual-gradient descent reach the same place starting from a plain standard chain-of-thought? We answer this with a head-to-head control that runs the optimiser in the opposite direction—initialised at the parent paper’s `standard_cot` prompt and descending on the same loss (Equation 1)—and

then pits the best optimised CoT against hand-written NoT at both the single-agent and the multi-stakeholder layers.

**Setup.** We optimise `standard_cot` on `claude-haiku-4-5` (which, unlike `gpt-5.4-nano`, retains failure-mode signal) under both the binary failure-mode loss and the continuous loss of Equation 1, training on seed-42 indices 0–29 and carrying forward whichever optimised prompt scores higher on held-out depth. All comparisons use the held-out scenarios (seed-42 indices 30–59), disjoint from training. At the single-agent layer, four conditions (NoT, the two optimised CoTs, and verbatim standard CoT) are generated on two generators (`claude-haiku-4-5`, `gpt-5.4-nano`) and coded by a cross-vendor judge pair (`claude-haiku-4-5` primary, `gpt-5.4-nano` secondary). At the multi-stakeholder layer we hold the parent paper’s five-round integration protocol and the moderator (`claude-sonnet-4-6`) fixed and vary only the agent prompt across three arms (NoT, best optimised CoT, plain standard CoT),  $n = 60$  debates per arm. Cliff’s  $\delta$  is signed so that positive favours NoT.

**Single-agent: NoT leads the best optimised CoT.** Even after optimisation, standard CoT does not reach NoT’s stakeholder coverage. Against the strongest optimised prompt, NoT retains Cliff’s  $\delta = +0.78$  (95% CI [+0.63, +0.91]) on stakeholder count on haiku and +0.07 ([-0.21, +0.34]) on nano, and +0.23 / +0.54 on uncertainty score (both CIs above 0). The optimised CoT closes the stakeholder-count gap on nano but never the uncertainty gap, at 2–5 $\times$  fewer tokens—it recovers breadth on the easier generator but not the explicit-uncertainty habit that NoT enforces by construction.

**Multi-stakeholder: NoT dominates the optimised CoT.** Table 5 reports the three-arm consensus result. NoT reaches full  $R4$  consensus on 52% of held-out scenarios versus 32% for the best optimised CoT (a +20pp gap, Fisher exact  $p = 0.041$ , two-proportion  $z$   $p = 0.026$ ) and leaves roughly four times fewer debates in unresolved structural rejection (8% vs 35%). Strikingly, optimising a CoT prompt for single-agent stakeholder breadth—the very axis the optimiser targets—*degrades* its behaviour once it is placed in the social protocol: the optimised-CoT arm is

the worst of the three on consensus and the most deadlocked.

**Honest caveat.** A plain, unoptimised standard CoT reaches the *highest* raw consensus of the three arms (63%, above NoT’s 52%). Our multi-stakeholder claim is therefore the specific one that NoT dominates the *optimised* CoT baseline, not that NoT maximises raw agreement. Higher raw consensus is not self-evidently better—the parent paper values agreement that survives genuine stakeholder tension rather than agreement reached by collapsing it.

**Where rejections land.** Coding every per-perspective *R4* vote (a cached judge over each debater’s final verdict) reframes the standard-CoT caveat as evidence for NoT. NoT’s *R4* rejections are both rare and role-appropriate: of the 5 per-perspective rejections across 126 NoT votes, 60% come from the External Advisor—the role explicitly charged with flagging principle violations—and only 20% from the Primary Decision Maker meant to drive toward a workable decision. The optimised CoT instead scatters 38 rejections near-uniformly across roles (normalised role entropy 0.98 vs. NoT’s 0.86), with the decider *itself* the modal rejector (42%): the role responsible for convergence most often blocks the synthesis it helped construct. NoT thus disagrees roughly  $7\times$  less often than the optimised CoT, and its dissent concentrates where principled objection belongs rather than scattering. (NoT’s narrative style left 12 of 126 verdicts without a clean binary token; these are counted conservatively as non-rejections, and NoT’s  $n = 5$  rejections make the share suggestive rather than a tight estimate.) The defensible conclusion here is narrow and clean: textual-gradient descent applied to a flat CoT does not reconstruct NoT at either layer, which is why the rest of this paper optimises *from* NoT rather than toward it.

## Limitations

**Single dataset.** All evaluations use DailyDilemmas (Chiu et al., 2025); generalisability to other ethical-reasoning benchmarks (ETHICS, Moral-Bench) is untested.

**Single optimiser model.** Textual gradients and prompt rewrites are produced by claude-sonnet-4-6. A different optimiser may produce different prompts; we do not separate the contribution of the optimiser model from the

| Arm           | <i>R4</i> consensus | 95% CI   | struct. reject |
|---------------|---------------------|----------|----------------|
| NoT           | 52%                 | [39, 64] | 8%             |
| optimised CoT | 32%                 | [21, 44] | 35%            |
| standard CoT  | 63%                 | [51, 74] | 17%            |

Table 5: Multi-stakeholder head-to-head on 30 held-out DailyDilemmas scenarios (2 generators, moderator claude-sonnet-4-6,  $n = 60$  debates per arm; only the agent prompt varies). NoT vs. optimised CoT: +20pp on full *R4* consensus (Wilson CIs shown), Fisher exact  $p = 0.041$ , and four times less unresolved structural rejection. Plain standard CoT reaches the highest raw consensus (caveat in text).

training-judge configuration.

**Narrow loss.** The optimisation loss targets stakeholder count and uncertainty score only. Max causal hops, action commitment, and trace coherence are out of loss; we report max-causal-hops deltas as a generalisation check (Table 3) but did not optimise against them.

**Sample-size variation.** Per-generator sample counts follow the parent study’s cost constraints ( $N = 5$  for nano, grok, sonnet;  $N = 20$  for haiku). The Cliff’s  $\delta$  CIs in Table 1 accommodate the variation, but the small- $N$  generators have wider intervals.

**Incomplete v1 baseline caches.** For nano and haiku, only 30 and 254 NoT-v1 judge cells (respectively) survived the parent paper’s cache rebuild, rather than the full 500 and 2000 originally generated. The Cliff’s  $\delta$  comparisons are still well-powered ( $\delta = -0.73$  for haiku with CI  $[-0.78, -0.68]$  on  $\geq 5 \times 10^5$  pairwise comparisons;  $\delta = +0.12$  for nano with CI  $[-0.15, +0.38]$ ), but the v1 mean estimates carry higher variance than the v2 and v3 ones. A future replication will regenerate the missing v1 cells.

**Generator overlap with judges.** The cross-family training judge (grok-4-1-fast-reasoning, Run B) and the cross-vendor evaluation third judge are the same model: it is the only non-Anthropic non-OpenAI judge available on our three-vendor panel. A fully orthogonal design would use a fourth vendor for the evaluation third judge; we do not have one. Within this constraint our claim is the narrower one: cross-family training on this three-vendor panel produces a prompt that survives cross-vendor evaluation strictly better than in-family training does.

**Within-family kappa paradox.** Six of eight

$\kappa$  rows in Table 4 are degenerate; we rely on the continuous stakeholder-count means (Table 2) on those rows. A larger eval pool that pushed binary marginals away from the extremes would let  $\kappa$  carry the analytical load directly.

**Pre-registered verdict label.** The pre-registration document declared a REPLICATES/PARTIAL/DOES NOT REPLICATE verdict based on a threshold of  $\kappa \geq 0.70$  on every generator. The continuous-metric agreement analysis in Section 5.2 was added post-registration to interpret the daggered  $\kappa$  rows that the pre-registration did not anticipate; the verdict labels themselves are not used in the paper’s claims, which are pegged to the per-generator effect sizes and cross-judge gaps directly.

**Action-commitment unanalysed.** The parent paper’s decision-extraction rubric was applied per cell during replication but is not analysed here. Whether NoT-v3 changes the model’s final action commitment is open.

**CoT-control scope.** The optimised-CoT control (Section 7) uses a narrower panel than the main replication—two generators (claude-haiku-4-5, gpt-5.4-nano) and the held-out 30-scenario split. Its role-concentration finding rests on NoT’s  $n = 5$  per-perspective  $R4$  rejections, so the modal-role share is directionally clear but statistically thin, and 12 NoT verdicts lacked a clean binary token (counted as non-rejections). We also define “materially undermined” indirectly via the rejecting role rather than via the integrator’s unaddressed-modification list, which was empty in almost every debate and so could not discriminate the arms.

## References

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## A Reproducibility

Pre-registration document: Guidance\_Documents/followup\_study\_design.md.  
Per-iteration optimisation artefacts (input prompt, batch outputs, judge codes, gradient, output prompt): divergence\_study\_outputs/ng\_iter\_{00..04}.json (Run A) and ng2\_iter\_{00..07}.json (Run B). Aggregated replication metrics: followup\_aggregate.json (v2) and followup\_v3\_aggregate.json (v3). Per-generator CSV summaries: followup\_per\_generator.csv and followup\_v3\_per\_generator.csv.

## B Verbatim Optimised Prompts

**NoT-v2 (in-family training; SHA-1 prefix 3fb7f75f6d; 2,579 characters).**

You are a thoughtful advisor reasoning through ethical dilemmas. Before giving your final answer, work through a structured analysis. Keep each section tight and purposeful.

**Who is deciding?** Name the decision-maker, their role, and the specific knowledge or constraints they bring to this choice.

**Who is affected?** Identify every person, group, or institution whose wellbeing, interests, or rights intersect this decision. Push past the central figures: include indirect parties, bystanders, background institutions, and anyone whose situation quietly changes as a result. For each, state precisely what they stand to gain, lose, or have changed—not that they are “affected,” but what specifically shifts for them. Prefer concrete specificity (“the clinic’s nursing staff who bear legal exposure”) over abstract categories. Aim for at least five distinct stakeholders with individuated stakes.

**What happens next?** For each available course of action, trace consequences at least two steps forward. Every stakeholder you named must appear somewhere in this analysis.

**What don’t we know?** For each projected future, surface specific uncertainties. For each course of action, name at least three distinct uncertainties—not generic hedges, but named gaps that could materially change the outcome. For each uncertainty, explain why it matters: what reversal or surprise would it produce?

**What should be done?** Commit to a specific decision. Explain why this path is preferable given the stakeholder impacts you

671 traced and the uncertainties you surfaced.  
 672 Name which uncertainties most threaten  
 673 your chosen path and state directly why you  
 674 are proceeding despite them.

675 *Additional guidance:* Precision over length.  
 676 Do not resolve tensions prematurely. Be  
 677 concrete about stakes. Explain the conse-  
 678 quence of being wrong, not merely that the  
 679 unknown exists.

## 680 NoT-v3 (cross-family training; SHA-1 prefix 681 a51ec242d5; 2,401 characters).

682 You are a thoughtful advisor helping people  
 683 navigate ethical dilemmas. Before commit-  
 684 ting to any recommendation, think carefully  
 685 and concisely through the following.

686 **Who is deciding and what do they stand**  
 687 **to gain or lose.** Characterize the decision-  
 688 maker's role, relevant knowledge, and per-  
 689 sonal stakes in one or two sentences.

690 **Who else is affected.** Cast a wide net. Start  
 691 with those directly executing or immediately  
 692 experiencing the decision, then move out-  
 693 ward: people affected one or two steps re-  
 694 moved, institutions and communities absorb-  
 695 ing indirect effects, and anyone whose fu-  
 696 ture options will be constrained by what hap-  
 697 pens now. For each distinct party, name their  
 698 concrete stake in a single sharp sentence.  
 699 Push past obvious stakeholders—someone  
 700 in the background or a future generation is al-  
 701 most always relevant. Aim to surface at least  
 702 six distinct parties; if you find more, include  
 703 them.

704 **What could go wrong or remain unknown.**  
 705 For every party you named and every re-  
 706 alistic course of action you are consider-  
 707 ing, state at least one specific uncertainty—  
 708 a hidden intention, an unpredictable reac-  
 709 tion, a missing fact, or a long-run effect that  
 710 cannot be resolved with available informa-  
 711 tion. Name the precise unknown, not merely  
 712 that uncertainty exists. No party should be  
 713 skipped. Where an uncertainty is especially  
 714 consequential—where resolving it would flip  
 715 the recommended action—say so explicitly.

716 **What the realistic options are and how**  
 717 **outcomes ripple.** For each plausible  
 718 course of action, trace the most decision-  
 719 relevant consequences forward. Focus on  
 720 how an outcome for one party reshapes out-  
 721 comes for others. Be selective: include  
 722 causal chains that change the analysis; skip  
 723 those that don't.

724 **The call.** Commit to a specific course  
 725 of action. Justify it by direct reference to  
 726 the stakeholders and uncertainties you sur-  
 727 faced. Name the uncertainties you are ac-  
 728 cepting and explain why the expected bene-  
 729 fits still justify those risks. Do not hedge into  
 730 vagueness—a qualified commitment is fine,  
 731 but the recommendation must be actionable.

732 Throughout, write concisely. One sharp sen-  
 733 tence per stakeholder stake. One precise un-  
 734 known per party. Avoid restating what earlier  
 735 reasoning already established.

## 736 C Training Loss Curves

737 Both runs trained on the seed-43 stratified 30-  
 738 scenario sample with batch size 10 and early-stop  
 739 after three consecutive iterations with  $< 5\%$  loss  
 740 reduction.

### 741 Run A (in-family, NoT-v2; early-stop iter 4).

| Iter       | $L$   | $\overline{sc}$ | $\overline{us}$ |
|------------|-------|-----------------|-----------------|
| 0 (NoT-v1) | 0.100 | 4.50            | 3.00            |
| 1          | 0.000 | 6.30            | 3.00            |
| 2          | 0.000 | 6.60            | 3.00            |
| 3          | 0.000 | 5.40            | 3.00            |
| 4          | 0.000 | 6.30            | 3.00            |

### 743 Run B (cross-family, NoT-v3; early-stop iter 7).

| Iter       | $L$   | $\overline{sc}$ | $\overline{us}$ |
|------------|-------|-----------------|-----------------|
| 0 (NoT-v1) | 0.100 | 4.50            | 2.90            |
| 1          | 0.000 | 7.70            | 3.00            |
| 2          | 0.000 | 6.30            | 3.00            |
| 3          | 0.100 | 4.60            | 2.90            |
| 4          | 0.000 | 6.10            | 3.00            |
| 5          | 0.000 | 5.90            | 3.00            |
| 6          | 0.000 | 5.70            | 3.00            |
| 7          | 0.000 | 5.70            | 3.00            |

745 **Drift control.** Both runs re-issued the NoT-v1  
 746 prompt on scenario dd\_32489 at the start and end  
 747 of training. Stakeholder counts: Run A  $6 \rightarrow 6$   
 748 ( $\Delta = 0$ ); Run B  $5 \rightarrow 6$  ( $\Delta = +1$ ). Both below  
 749 the pre-registered  $\Delta = 0.5$  threshold for compute-  
 750 drift confounding (Run B's  $+1$  is within rounding  
 751 of the threshold on a single-cell measurement; the  
 752 four-generator replication shows no compute-drift  
 753 signature).